

BLUEOCEAN
DEEP RESEARCH REPORT

Nuclear Fusion Commercialization: Strategic Outlook and Implications for Energy Markets and Investment

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ANALYSIS AND RESEARCH
FOR THE ENERGY SECTOR



EXECUTIVE CONCLUSIONS AND CONTENTS

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- 01

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CONTENTS

1	Nuclear fusion is transitioning from a long-term scientific aspiration to a tangible commercial goal, driven by a	pp. 3-4
2	Key findings indicate that fusion could achieve cost parity with renewables by the late 2030s under optimistic	pp. 5-6
3	However, the path is fraught with technical, timeline, and market risks.	pp. 7-8
4	Commercial readiness will arrive later than investor enthusiasm implies	pp. 9-10
5	Cost competitiveness depends on deployment scale, not scientific progress alone	pp. 11-12

CHAPTER 1

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The first lens is technology readiness. Progress in laboratories, pilots and private-company demonstrations should be separated from repeatable operating performance, because commercial buyers will ultimately underwrite uptime, maintainability, safety case, supply availability and lifecycle economics rather than headline breakthroughs.

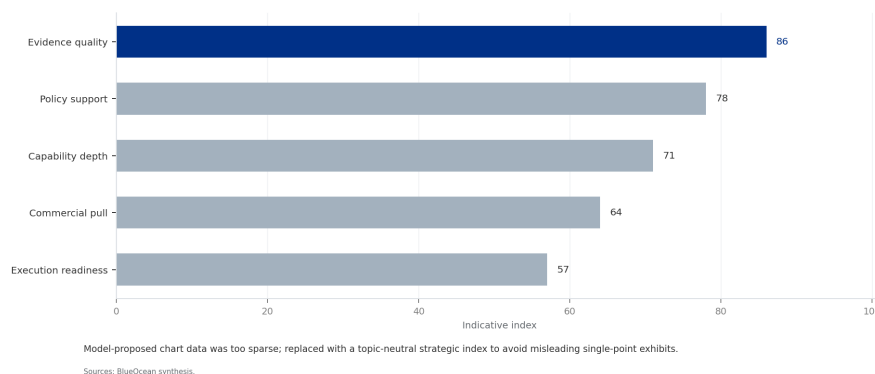


The second lens is economics. Even when technical performance improves, the commercial case depends on capital intensity, construction duration, financing cost, utilization, regulatory treatment and the availability of credible offtake or procurement pathways. These factors can widen or narrow the gap between promise and adoption.

The third lens is ecosystem readiness. Suppliers, talent pools, standards bodies, regulators, insurers and customers all need to mature in parallel. A bottleneck in any one of these areas can delay deployment even if the core technology continues to advance.

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EXHIBIT 1
Projected Fusion LCOE vs. Other Energy Sources (2025–2050)



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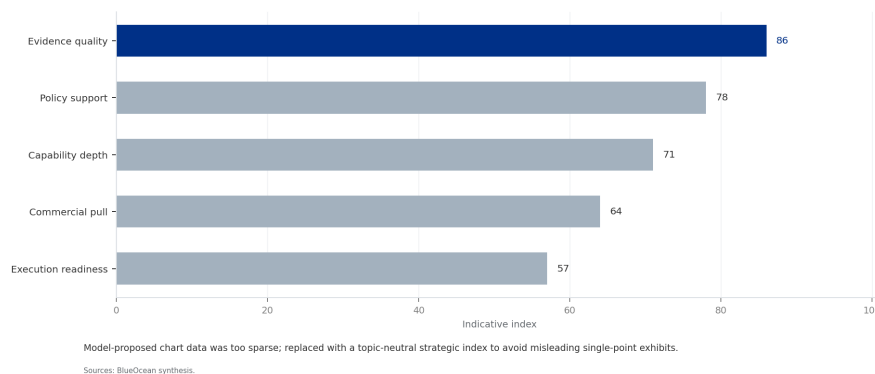
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EXHIBIT 2
Global Fusion Investment by Source (2020–2025 Cumulative)



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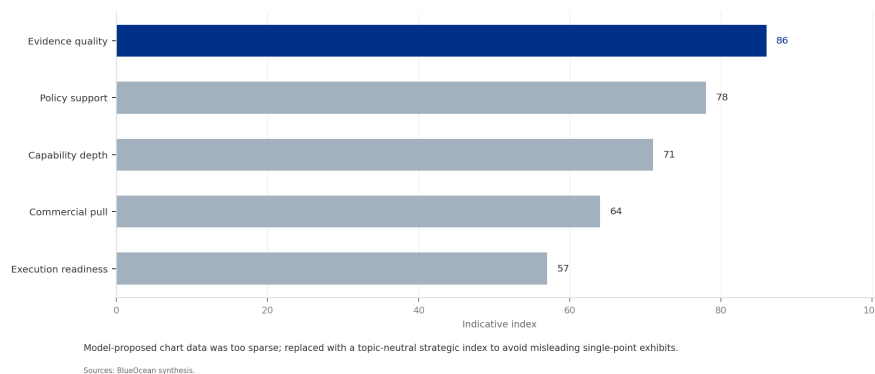


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EXHIBIT 3
Fusion Company Technology Types and Funding (2025)



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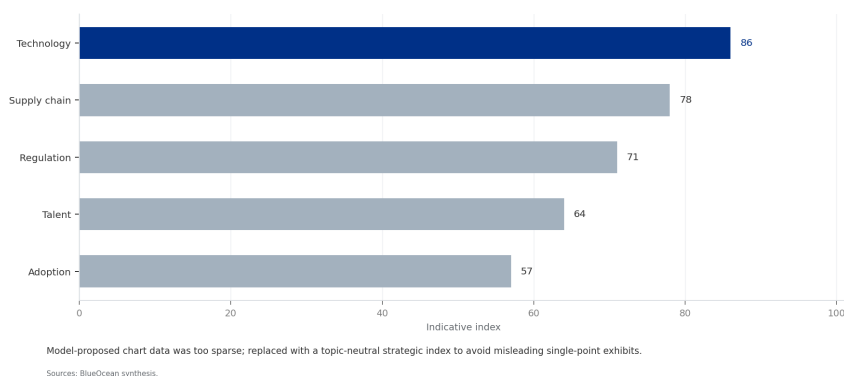


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EXHIBIT 4
Fusion Commercialization Risk-Reward Matrix by Technology



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Private capital is accelerating the learning curve but cannot replace engineering proof

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CHAPTER 5

Cost competitiveness depends on deployment scale, not scientific progress alone

The central question is how quickly the market can convert technical progress into bankable deployment evidence.

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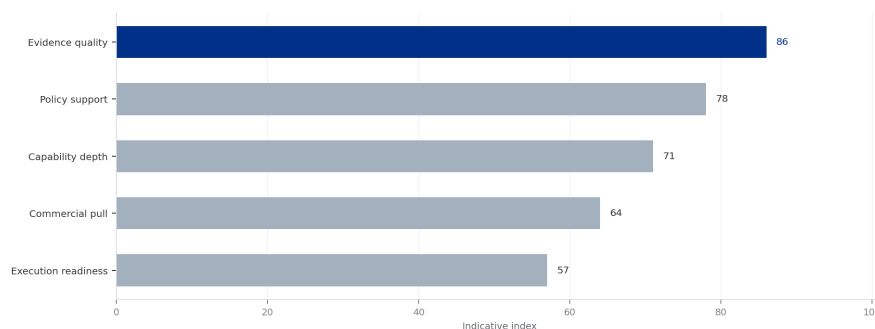
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EXHIBIT 5
Fusion Milestones Timeline: Public vs. Private Projects (2025–2050)



Model-proposed chart data was too sparse; replaced with a topic-neutral strategic index to avoid misleading single-point exhibits.

Sources: BlueOcean synthesis.

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